

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2025
 Programme: BE Full Marks : 100
 Course: Probability and Statistics (New) Pass Marks : 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) The life of two models of Ceiling Fans in a recent survey is given below. 8

Life (in Years)	Number of Ceiling Fan	
	Model 1	Model 2
0 – 2	5	2
2 – 4	16	7
4 – 6	13	12
6 – 8	7	19
8 – 10	5	9
10 – 12	4	1

- i. What is the average life of each model of Ceiling Fans?
 ii. Which model of Ceiling fan has greater uniformity?
- b) In a class of 75 students, 15 students were considered to be very intelligent, 45 are medium and rest are below average. The probability that a very intelligent student fail in an examination is 0.005, the medium is 0.05 and the below average is 0.15. If a student passed in a examination, what is the probability that he/she is below average?
2. a) In a certain city a survey of 600 families about the fuel for cooking revealed that 230 use L.P. Gas and 175 use Induction and 40 use both. Then find the probability that a family selected who use
 i. At least one of the fuel
 ii. Neither of the fuel
 iii. Exactly one type of fuel
- b) A coin is tossed until a head appears. What is the expectation number of tosses required? 7

OR

A random variable X has the following probability function then

X	2	3	4	5	6	7
P(X=x)	0.1	k	0.2	2k	0.3	k

- i. Find the value of k?
 ii. Find E(X) and V(X).
 iii. Find the distribution function of X.
3. a) An aircraft has 116 seats that airlines has found. From long experience that on average 2.5 % of the people have bought the tickets for the flight do not arrive for the flight. The airlines sell 120 tickets for a flight to Pokhara then calculate the probability that
 i. More than 118 people arrive for the flight?
 ii. There are empty seats on the flight?
- b) In an examination 15% of the students got first class marks above 60 and while 40% failed marks below 40. Assuming the marks to be normally distributed, estimate the mean and standard deviation. Also, find its coefficient of variation.
4. a) Suppose that X is a random variable whose probability density function is given by, $f(x) = C(4x - 2x^2)$, $0 < x < 2$, $0 < y < 1$
 i. Determine the value of C.
 ii. Determine the cumulative Probability of X.
 iii. Find the Marginal density function of X and Y.
 iv. Find the conditional density function of X given Y=y.
- b) Define Estimation. A hotel manager in Kathmandu wants to know the hotel's average daily registration. The following table presents the number of guests registered for 14 randomly selected days. Calculate the sample mean, standard error, and 95% confidence interval of the population mean.

61	57	60	57	54	58	55
52	65	40	64	70	22	56

OR

Define point and interval estimation. A certain Cinema Hall sampled 55 people who viewed Hero Number One and asked them whether they planned to see it again. Only 10 of them believed the film was worthy of a second look.

- i. Estimate the standard error of the proportion of movie viewers who will view the film a second time.

- ii. Construct a 95% confidence interval for this proportion.
5. a) A population consists of the values 1, 2, 3, 6, 8. Prove that the sample mean of size 3 is an unbiased estimate of the population mean (without replacement). 7

OR

Two independent random samples of 70 boys and 55 girls from a normal population were drawn, and their mean weights were 62 kg and 46 kg, with variances of 16 and 12, respectively. Construct a 95% confidence interval for the difference between the mean weight of boys and girls.

- b) A random sample of 400 males and 600 females was asked whether they would like to have an airport near their residence. 200 men and 325 women were in favor of the proposal. Test the hypothesis that the proportion of men and women is in favor of the proposal at 5% level of significance 8
6. a) A principal of the college thought that memory power training is better for the learning process of students, so memory power training was launched. The memory capacity of the students was noted before and after training. The data is recorded as: 7

Is the training effective at 5% level of significance?

Before Training	56	42	36	47	49	42	60	72	63
After training	147	125	118	128	145	140	155	160	149

- b) The following table gives the age X and blood pressure Y of 10 women. 8

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	149	150

- i. Find the correlation coefficient between age and pressure.
- ii. Fit a regression equation of X on Y.
7. Write short notes on: (Any two) 2×5
- a) Probability and Non-Probability Sampling
- b) Differentiate between Cluster Sampling and Stratified Sampling Methods
- c) Define Type I and Type II errors in Hypothesis Testing